

1. The central dogma of molecular biology tells us that information is passed from 1 point
- ☐ DNA to protein to RNA
 - ☐ RNA to DNA to protein
 - ☐ DNA to RNA to methylation to protein
 - ☒ DNA to RNA to protein
2. Which of the following is one of the major drivers of the sequencing revolution that began after 2008? 1 point
- ☐ Decreased computational analysis time
 - ☐ Increased sample collection
 - ☐ Improved Sanger sequencing
 - ☒ Decreased cost of sequencing
3. Which of the following is an exclusive characteristic of genomics compared to traditional biology? 1 point
- ☒ Studies considering the entire genome
 - ☐ Low throughput experiments
 - ☐ Targeted studies of one or a few genes
 - ☐ Clever experimental design
4. Genomic data science involves techniques from which of these disciplines? 1 point
- ☐ Molecular Biology
 - ☒ All of the these options
 - ☐ Statistics
 - ☐ Computer Science
5. Which of the following is an activity that genomic data scientists do not perform? 1 point
- ☒ Pipetting
 - ☐ Integrative genomics
 - ☐ Statistics and machine learning
 - ☐ Population genomics
6. Which of these is not one of the DNA nucleic acids? 1 point
- ☒ Alanine
 - ☐ Adenine
 - ☐ Cytosine
 - ☐ Thymine

7. Transcription is a process that converts DNA to

1 point

- ☐ polymerases
- ☒ RNA
- ☐ genes
- ☐ ribosomes

8. The cost to sequence a human genome today, in U.S. dollars, is approximately

1 point

- ☒ \$1000
- ☐ None of these options
- ☐ \$3 billion
- ☐ \$30 million

9. DNA encodes instructions for

1 point

- ☒ Producing all the proteins that a person requires for life
- ☐ Helping us digest food
- ☐ Enveloping viruses that infect a cell
- ☐ Regulating body temperature

10. One major difference between humans and bacteria is

1 point

- ☒ Human cells contain separate organelles called mitochondria, and bacterial cells do not.
- ☐ The human genome is made of DNA, while bacteria are made of RNA.
- ☐ Human genes are first transcribed to RNA, while bacterial genes are not.
- ☐ Human proteins are made of combinations of 20 amino acids, while bacterial proteins use a smaller set of 12 amino acids.